ModelIt!

Teacher's Guide

The Energy Roller Coaster: Where Does Speed Come From?

How Energy Transforms, Transfers, and Gets Lost Along the Way

Quick Reference

Grade Level	Grade 04L2
NGSS Standard	4-PS3-1, 4-PS3-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 4-PS3-1, 4-PS3-4

Use evidence to construct an explanation relating the speed of an object to the energy of that object. Apply scientific ideas to design, test, and refine a device that converts energy from one form to another.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model with five components showing how hill height and friction interact to determine kinetic energy through energy transformation and heat loss.
Disciplinary Core Idea	**PS3.A Definitions of Energy / PS3.B Conservation of Energy** — Energy can be moved from place to place and can be transferred between objects. Energy is conserved — when a moving object slows down, its energy is transferred to heat through friction.
Crosscutting Concept	**Energy and Matter** — Students trace energy through a system, identifying how it transforms from potential to kinetic to heat, demonstrating conservation while explaining why motion decreases.

CAST Testing Alignment

MULTIPLE CHOICE:

A marble rolls down a tall ramp on a smooth surface and then on a rough surface. It travels farther on the smooth surface. Which best explains why?

CONSTRUCTED RESPONSE:

Using what you know about energy conservation and friction, explain why a roller coaster eventually stops even though energy can never be destroyed. Use the words 'potential energy,' 'kinetic energy,' and 'heat loss' in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Energy transformation is one of the most powerful ideas in science. Every ride at a theme park, every car on the road, and every ball you throw demonstrates how energy changes form. Understanding friction's role in energy transformation explains why perpetual motion machines are impossible and why engineers work so hard to reduce friction in machines.

Energy Transformation on a Roller Coaster

A roller coaster is a perfect laboratory for studying energy transformation. At the top of the first hill, the coaster has maximum potential energy (PE) — energy stored because of its height. As it descends, PE converts to kinetic energy (KE) — the energy of motion. At the very bottom, nearly all PE has become KE, and the coaster is moving at its fastest speed. As it climbs the next hill, KE converts back to PE. This back-and-forth conversion would continue forever if not for one thing: friction.

Friction and Heat Loss

Friction occurs wherever two surfaces slide against each other. On a roller coaster, friction happens between the wheels and the track, and between the coaster and the air (air resistance). Friction transforms kinetic energy into thermal energy (heat). You can feel this effect by rubbing your hands together quickly — they get warm because friction converts your motion energy into heat. On a coaster, this heat radiates into the air and is lost from the motion system forever.

The Law of Conservation of Energy

The total amount of energy in a closed system never changes — this is the Law of Conservation of Energy. When a roller coaster slows down, energy has not been destroyed. It has been transformed into heat, sound, and vibration. If you could measure ALL the energy (kinetic + potential + heat + sound), the total would equal the original potential energy at the top of the first hill. Engineers cannot eliminate friction, but they can reduce it with smooth surfaces, lubricated wheels, and aerodynamic designs.

Real-World Engineering Applications

Understanding energy and friction is critical for engineers in many fields. Car engineers design sleek shapes to reduce air friction and improve fuel efficiency. Roller coaster designers calculate exactly how much energy is lost to friction to ensure the car makes it through every loop and hill. Space engineers use friction (drag) to slow spacecraft when entering an atmosphere. Even ice skaters know that smoother ice means less friction and faster spins.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Hill Height and Track Friction as external components (design choices) and Potential Energy, Kinetic Energy, and Heat Loss as internal components that respond to the externals.

E **Establish Relationships**

Students determine that Hill Height positively affects PE, PE positively affects KE, Track Friction positively affects Heat Loss, and Heat Loss negatively affects KE — creating a system with competing influences.

V **Visualize & Model**

Students build a five-component model showing how energy flows from height to potential energy to kinetic energy, while friction diverts energy into heat loss that reduces speed.

E **Evaluate Outcomes**

Students run four scenarios (smooth tall, rough tall, smooth short, worst case) to observe how changing two external inputs creates different outcomes in the system.

R **Revise & Extend**

Students add loop size or air resistance to explore why some coaster designs fail — a loop requires minimum KE to complete, so too much friction can make the coaster fall at the top of a loop.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Who remembers our first roller coaster lesson? Today we are going DEEPER — we are going to figure out why every roller coaster eventually stops."*

Watch For

Show a quick video of a roller coaster slowing to a stop. Ask: Where did all that speed go?

Slide 2: Learning Objectives

3 min

Say: *"Today we have FIVE system parts to work with instead of three — we are leveling up our modeling skills!"*

Watch For

Have students read objectives aloud. Review PE and KE from L1, then introduce friction and heat loss.

Slide 3: Big Question

3 min

Say: *"Here is the puzzle: scientists say energy can NEVER be destroyed. But the coaster stops. So where does the energy GO?"*

Watch For

Have students turn and talk: If energy cannot be destroyed, why does the coaster eventually stop? Write predictions.

Slide 4: LEVER Framework

2 min

Say: *"This model is more complex than our first one. We have two things we control and three things that respond."*

Watch For

Preview the five components. Explain that having two external components means we can test combinations.

Slide 5: Activity 1: Components

8 min

Say: *"Which of these five parts do WE control, and which ones respond on their own?"*

Watch For

Guide sorting of five components. Discuss why Hill Height and Track Friction are external (design choices) while PE, KE, and Heat Loss are internal (they respond automatically).

Slide 6: Activity 2: Connections

10 min

Say: *"Here is where it gets interesting — some arrows are POSITIVE and some are NEGATIVE. What does a negative arrow mean?"*

Watch For

Help students draw all four relationships. Emphasize the negative arrow from Heat Loss to KE — this is the key insight.

Slide 7: Activity 3: Simulation

12 min

Say: *"Watch what happens when we change friction. Keep your eyes on the KE and Heat Loss lines!"*

Watch For

Run scenarios in order. Have students record observations in their activity pack. Focus on the smooth vs. rough comparison.

Slide 8: Discoveries

5 min

Say: *"Now we know the answer — energy is CONSERVED but TRANSFORMED. Friction turns motion energy into heat!"*

Watch For

Have students rub hands together to feel friction heat. Draw the complete energy flow on the board.

Slide 9: STEM Challenge

12 min

Say: *"You are engineers now. Your mission: design a track that fights friction and keeps the marble fast!"*

Watch For

Distribute materials (ramps, foil, wax paper, sandpaper, marbles). Groups test surfaces and record distances.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Hill Height, Track Friction
- **INTERNAL:** Potential Energy, Kinetic Energy, Heat Loss

Relationships

- **Hill Height to Potential Energy:** POSITIVE (+) — A taller hill lifts the coaster higher above the ground, storing more gravitational potential energy. Doubling the height roughly doubles the stored energy available for conversion to motion.
- **Potential Energy to Kinetic Energy:** POSITIVE (+) — As the coaster descends, stored potential energy converts into kinetic energy (motion). More PE at the top means more KE at the bottom — this is direct energy transformation.
- **Track Friction to Heat Loss:** POSITIVE (+) — A rougher track surface creates more friction between the wheels and the rails. More friction means more kinetic energy gets transformed into heat energy, which escapes into the surrounding air.
- **Heat Loss to Kinetic Energy:** NEGATIVE (-) — Energy lost as heat is energy that is NOT available for motion. When Heat Loss increases, Kinetic Energy decreases because the total energy is conserved — more going to heat means less going to speed.

Key Discoveries

- Taller hills give the coaster more potential energy at the top, which converts to more kinetic energy (speed) at the bottom
- Friction transforms kinetic energy into heat — the energy does not disappear, it just changes into a form that does not help the coaster move
- When friction is high, more energy is lost as heat, so less energy is available for motion — the coaster slows down
- Energy is CONSERVED (never created or destroyed), but it can change forms — and once it becomes heat from friction, it cannot easily change back to motion

Multiple Choice Post-Assessment Answer Key

1. A roller coaster starts at the top of a tall hill and rolls down to the bottom. What happens to its potential energy and kinetic energy during the ride down?

Correct: B) Potential energy decreases while kinetic energy increases

That is right! As the coaster rolls downhill, its stored potential energy transforms into kinetic energy. The higher the starting hill, the more kinetic energy the coaster gains at the bottom.

2. In the model, Hill Height and Track Friction are external components. Why are they external?

Correct: C) Because they are design choices we can control before the ride starts

Correct! External components are things we can set or control. Engineers decide how tall to build the hill and what surface material to use for the track before the coaster starts moving.

3. The model shows that when Track Friction increases, Heat Loss increases, and Kinetic Energy decreases. What does this tell us about friction?

Correct: C) Friction transforms kinetic energy into heat, leaving less energy for motion

Exactly! Friction transforms motion energy into heat energy. The energy is not destroyed, it just changes form. With more energy going to heat, less is available for speed.

4. A student tests two roller coasters. Coaster A has a tall hill and a smooth track. Coaster B has the same tall hill but a rough track. Which coaster will be faster at the bottom, and why?

Correct: C) Coaster A, because less friction means less energy is lost to heat, so more energy stays as kinetic energy

Right! Both coasters start with the same potential energy, but Coaster A's smooth track loses less energy to heat. More energy stays as kinetic energy, so Coaster A is faster at the bottom.

5. The relationship between Heat Loss and Kinetic Energy is NEGATIVE. What does a negative relationship mean in this model?

Correct: B) When Heat Loss increases, Kinetic Energy decreases because energy going to heat cannot also be used for motion

That is right! A negative relationship means when one goes up, the other goes down. More energy lost as heat means less energy available for speed. Energy is conserved, so it can only be in one form at a time.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"The roller coaster runs out of energy"	Energy is NEVER used up or destroyed. When a coaster slows down, its kinetic energy has been transformed into heat energy through friction and sound energy from the rumbling. The total energy in the universe stays the same — it has just changed into forms that cannot push the coaster forward anymore.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Friction only happens on rough surfaces"	ALL surfaces have friction, even ones that look perfectly smooth. Under a microscope, every surface has tiny bumps and ridges that catch on each other. Engineers can reduce friction but never eliminate it completely. Even ice has friction — that is why ice skaters eventually slow down.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"A taller second hill would make the coaster go faster"	The coaster can NEVER go higher than its starting hill (without an engine boost) because friction has already converted some energy to heat. Each hill on a coaster must be shorter than the previous one. If a hill were taller, the coaster would not have enough energy to make it over the top.